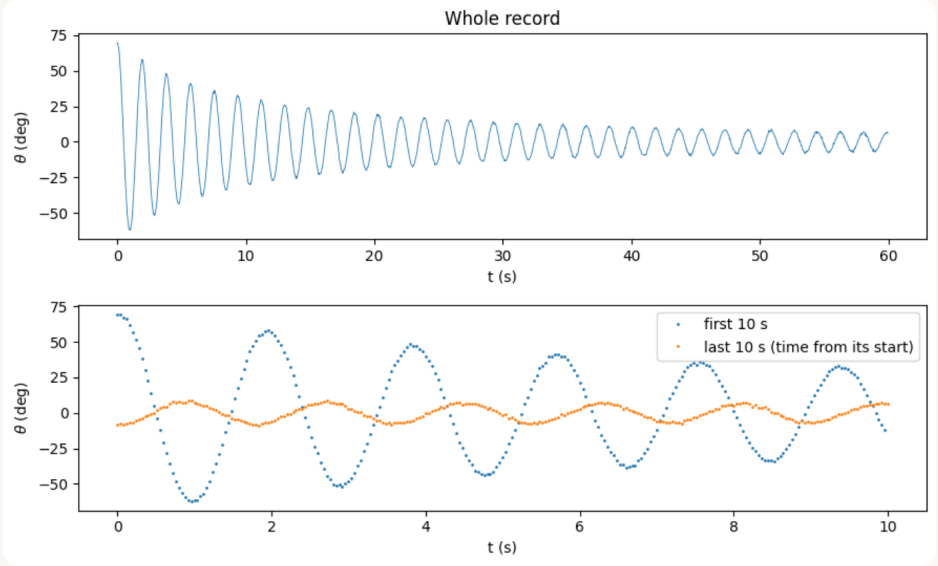
PHYS7101 · EXERCISE 1 FEEDBACK

What the pendulum was hiding

How we measure g, and how we decide whether to believe it

Same data, three models, three values of g



1500 tracked angles over 60 s, σ = 0.5°, released from about 70°.

Tasks 1–5 built the tests that judge a model. Each showed where the model fell short.

g is only as good as the model it is fitted with.

SMALL-ANGLE

g = 9.527

$\chi^2/\nu = 116$

→

NONLINEAR, RK4

g = 9.8344

$\chi^2/\nu = 27$

→

EXTENSION

g = ?

What is still missing?

The brief asked two questions

“Measure g — and decide how far you trust your answer.”

HANDS UP

What g did your Task 3.2 fit give, and what σ?

Each fit was better, and still wrong

3.1 SMALL-ANGLE FIT

9.527

$\chi^2/\nu = 116$

3.2 RK4, LINEAR DAMPING

9.8344

$\pm$ 0.0007, $\chi^2/\nu = 27$

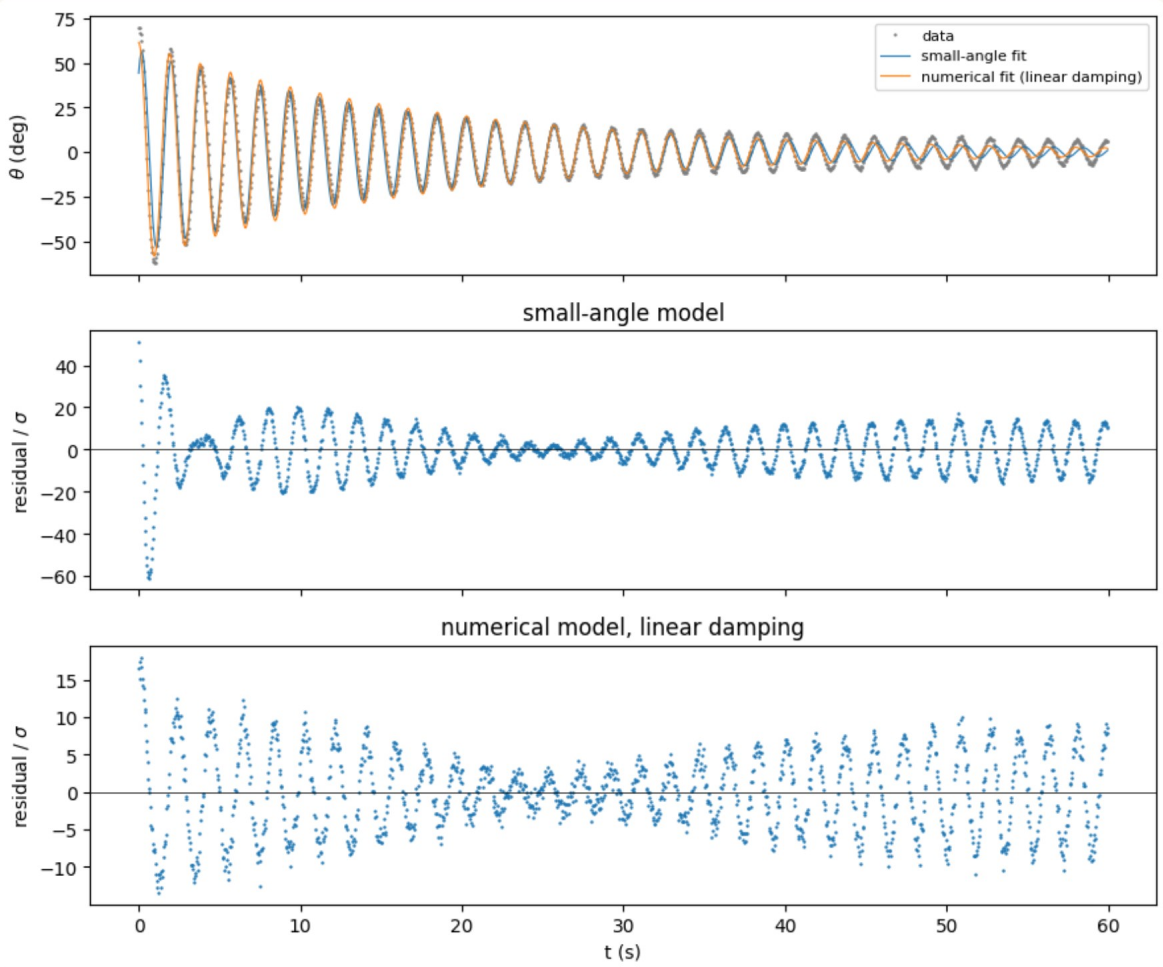
LOCAL G, LANCASTER

9.8141

WGS84 formula, 54.0° N

9.8344 is 30σ from 9.8141. A small σ is not a right answer.

The residuals had a pattern



Noise would scatter by ±1σ with no shape.

Both fits leave an **oscillating pattern**, many σ in size and **largest at the start**.

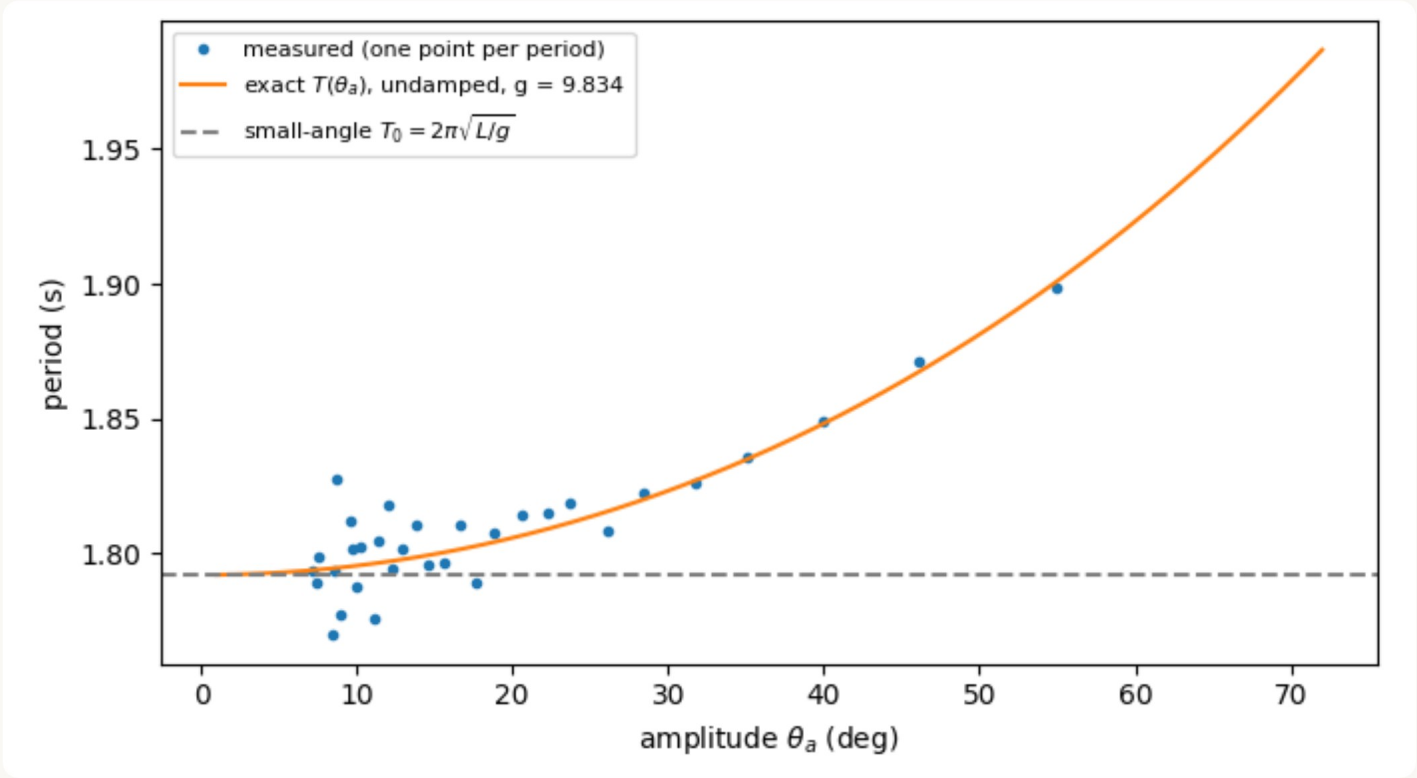
The eye sees it before χ^2/ν does.

PAUSE

$\chi^2/\nu = 27$ with $\nu \approx 1500$. Name three things that could cause it.

Then sort them: the data, the code, or the model?

The pendulum is not linear



Period falls from **1.90 s at 55°** to **1.79 s at 7°**.

It follows the exact curve, not the constant T_0.

ellipk takes m = k². Passing k gives a curve 11% high.

Three clues pointed at the model

RELEASE ANGLE · TASK 3.2

61° vs 70°

The fit puts θ_0 at 61.4°. The first frame reads 69.6°.

DAMPING · THE EXEMPLAR'S EXTENSION

2.5×

γ from the first 30 s is 0.125 s^{-1}; from the last 30 s, 0.049 s^{-1}.

VELOCITY · TASK 5.1

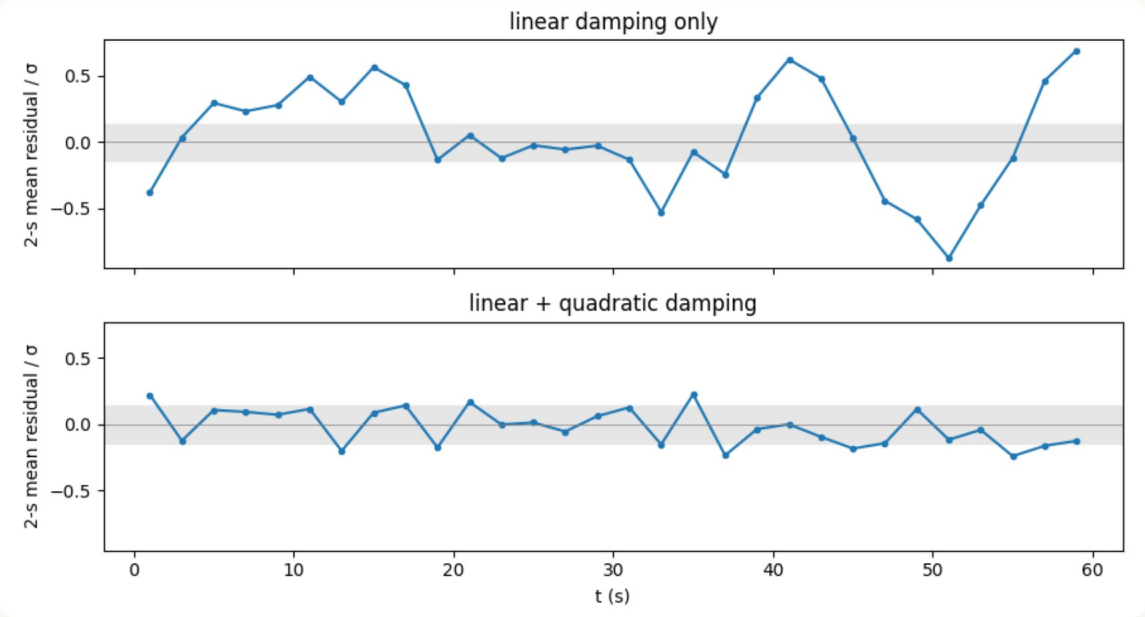
12.1 vs 8.8 °/s

Expected from noise: $\sigma/(\sqrt{2}\,\Delta t)$ = 8.8 °/s. The excess is 37%.

Damping that is **stronger when the bob moves fast**: that is air drag.

One extra term fixes every symptom

$$\ddot{\theta} + \gamma\dot{\theta} + \mathbf{c}\,\dot{\theta}|\dot{\theta}| + (g/L)\sin\theta = 0$$



	Linear	+ drag
χ^2/ν	26.9	1.02
θ_0	61.4°	70.0°
ω scatter	12.1 °/s	8.9 °/s
g (m/s²)	9.8344	9.8137
vs local g	+30σ	−0.7σ
Below 1°	76 s	189 s

You did not need to find the drag for an A

What earned credit was asking **“what is the 30σ?”** and testing an answer.

Fit in windows: does g or γ drift?	Solver-step check: is the 30σ numerical?	Small-angle refit of the tail
Moving-window velocities	Energy balance	Adding a drag term

Each counts when it is motivated by Tasks 1–5 and ends in a conclusion.

Five slips we saw often

What we saw	The fix
Large χ^2/ν blamed on σ, jitter or a bug	Judge it against 1. This large, the model is incomplete.
The 30σ gap stated, not explained	Quote the gap in σ, then ask what physics is missing.
12.1 °/s read as noise	Expected $\sigma/(\sqrt{2}\,\Delta t)$ = 8.8 °/s. The excess is the model.
Local g of 9.81 or 9.80665	Cite a source for 54.0° N: about 9.814.
Checks that could not fail	Test where error is largest: undamped, 70°, 60 s.

A strong statement answers five questions

Question	In the exemplar
1. What is g, with σ?	9.8137 ± 0.0005, 0.7σ from local g (cited)
2. Is it linear?	No: period 1.90 s at 55° to 1.79 s at 7°
3. Can the solver be trusted?	Energy to 2×10^{-5}, order 4, h/2 at $6\times10^{-5}\sigma$
4. What did the extension show?	Three symptoms, one missing term; σ checked by a pull study
5. What next?	Measure the drag directly, with a different bob shape

Didn’t finish? Answer them for the work you did: say why there is no extension and what you would do next. The statement is marked on the discussion, not on how far you got.

ACTIVITY

Read the exemplar statement. Tick off the five questions.

Then: which sentence does the most work, and why?

Three habits for Exercise 2

Habit	In Exercise 1	In Exercise 2
1. Ask what noise alone would give	$\chi^2/\nu = 27$, not 1. σ was measured, so the model is missing something	Are your values independent? What does chance alone give?
2. Quote every comparison with its uncertainty	“30σ from local g”, not “0.2% off”	A difference with its interval, not just “bigger”
3. Run each check where errors are largest	h/2 test: undamped, from 70°, over 60 s	Your artefact rule on the people with the most extra beats

Monday: the in-class assessment is a short conversation about your Exercise 2 work. Bring your notebook.

Exercise 2, Task 3.2 now asks for the spectrogram first. Work you have already done still counts.

Exercise 1 marks and feedback: on Moodle next week.